
BASIC VULNERABILITY ASSESSMENT REPORT

Prepared By:

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Environment:

Kali Linux (WSL) – Local Testing Lab

Date:

2026

1. INTRODUCTION

This assessment was performed in a controlled lab environment to understand the basic workflow of vulnerability assessment and security testing. The lab focused on setting up a testing environment, scanning systems for vulnerabilities, documenting findings, performing basic risk assessment, and preparing a professional security report.

All activities were performed strictly for educational and defensive security purposes using intentionally vulnerable or locally configured systems.

2. SETUP TESTING ENVIRONMENT

The first step of the lab was preparing a testing environment for vulnerability assessment activities.

Kali Linux was installed using Windows Subsystem for Linux (WSL). The environment was used as the main penetration testing workstation. Several security tools available in Kali Linux were verified successfully, including Nmap, Nikto, Metasploit Framework, and OpenVAS/GVM.

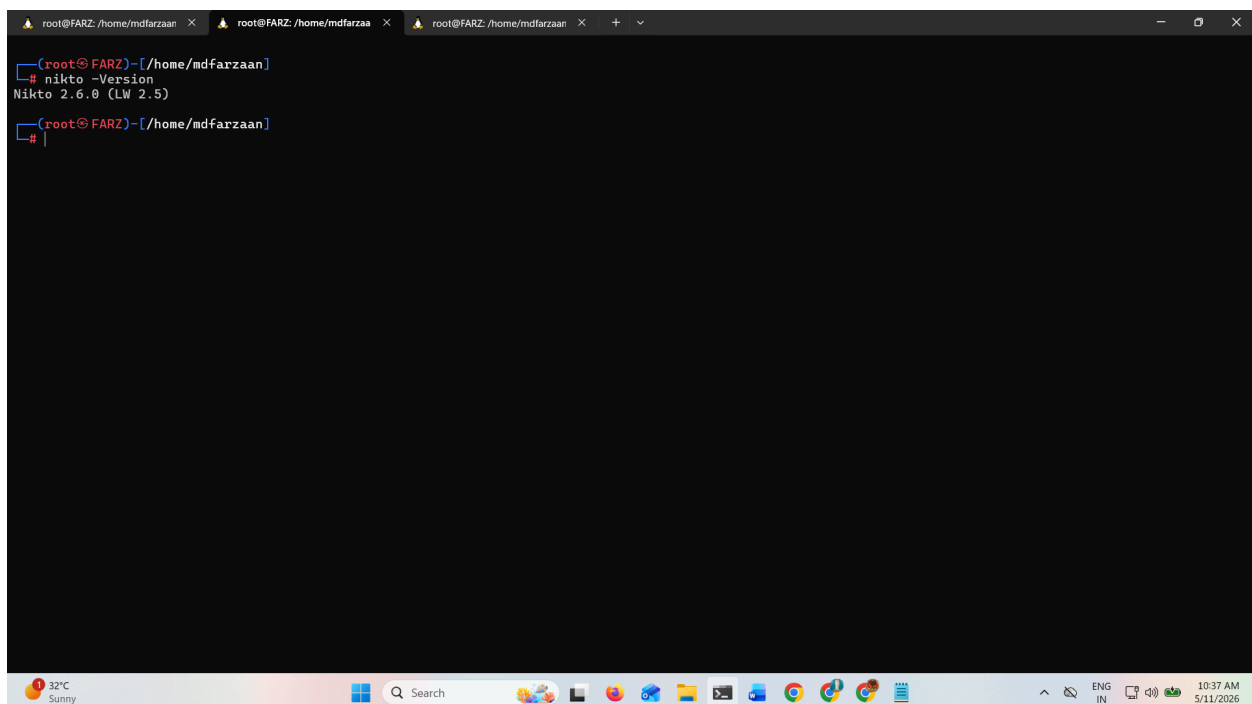
Virtualization concepts using VirtualBox were also studied to understand how isolated vulnerable virtual machines such as Metasploitable can be hosted safely for practice purposes.

The lab environment allowed testing activities to be performed without affecting external systems.

Tools Used:

- Kali Linux (WSL)
- VirtualBox
- GitHub Resources

Evidence:



```
root@FARZ: /home/mdfarzaan x root@FARZ: /home/mdfarzaa x root@FARZ: /home/mdfarzaan x + - x
(root@FARZ)-[/home/mdfarzaan]
# nikto -Version
Nikto 2.6.0 (LW 2.5)
(root@FARZ)-[/home/mdfarzaan]
# |
```

The screenshot shows a Windows taskbar at the bottom with a weather widget indicating 32°C and Sunny. The taskbar includes icons for Search, File Explorer, Microsoft Edge, and other applications. The terminal window has three tabs, all showing the same directory: root@FARZ: /home/mdfarzaan. The active terminal shows the command 'nikto -Version' being executed, resulting in the output 'Nikto 2.6.0 (LW 2.5)'. The system clock in the bottom right corner shows 10:37 AM on 5/11/2026.

```
root@FARZ: /home/mdfarzaan x root@FARZ: /home/mdfarzaan x root@FARZ: /home/mdfarzaa x + v
(root@FARZ) - [/home/mdfarzaan]
# nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.5.5 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(root@FARZ) - [/home/mdfarzaan]
#
```

```
root@FARZ: /home/mdfarzaan x root@FARZ: /home/mdfarzaa x root@FARZ: /home/mdfarzaan x + v
(root@FARZ) - [/home/mdfarzaan]
# nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-11 10:44 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
5432/tcp  open  postgresql   PostgreSQL DB 18.0 - 18.2

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.31 seconds

(root@FARZ) - [/home/mdfarzaan]
#
```

The image shows a web browser window displaying the OpenVAS login page. The browser's address bar shows the URL `127.0.0.1:9392/login`. The page has a green background with a white login form in the center. A warning message at the top states: "Warning: Connection unencrypted. The connection to this web application is not encrypted, allowing anyone listening to the traffic to steal your credentials. Please configure a TLS certificate for the HTTPS service or ask your administrator to do so as soon as possible." The login form includes fields for "Username" and "Password", a "Sign in" button, and the "OPENVAS COMMUNITY EDITION" logo. Below the browser window, a terminal window is open, showing the output of several commands: `whoami` (root), `pwd` (/home/mdfarzaan), `uname -a` (Linux FARZ 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux), and `ip a` (showing network interfaces lo, eth0, and docker0).

Warning: Connection unencrypted

The connection to this web application is not encrypted, allowing anyone listening to the traffic to steal your credentials.

Please configure a TLS certificate for the HTTPS service or ask your administrator to do so as soon as possible.

Sign in to your account

Username

Password

Sign in

OPENVAS
COMMUNITY
EDITION

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```

root@FARZ: /home/mdfarzaan
# whoami
root

root@FARZ: /home/mdfarzaan
# pwd
/home/mdfarzaan

root@FARZ: /home/mdfarzaan
# uname -a
Linux FARZ 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux

root@FARZ: /home/mdfarzaan
# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host proto kernel_l1
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1420 qdisc mq state UP group default qlen 1000
   link/ether 00:15:5d:2c:70:d1 brd ff:ff:ff:ff:ff:ff
   inet 172.26.106.18/20 brd 172.26.111.255 scope global eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::215:5dff:fe2c:70d1/64 scope link proto kernel_l1
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 6e:25:0f:23:45:11 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

root@FARZ: /home/mdfarzaan
#

```

3. VULNERABILITY SCANNING

Vulnerability scanning activities were performed using OpenVAS/GVM, Nmap, and Nikto.

OpenVAS/GVM was configured within the Kali Linux environment to understand vulnerability management workflows. The Greenbone web interface was successfully accessed, and service initialization was demonstrated.

Nmap was used to perform service detection and identify open ports on the target environment.

Command Used:

```
nmap -sV 127.0.0.1
```

Nikto was used to scan the local web service for insecure configurations and outdated components.

Command Used:

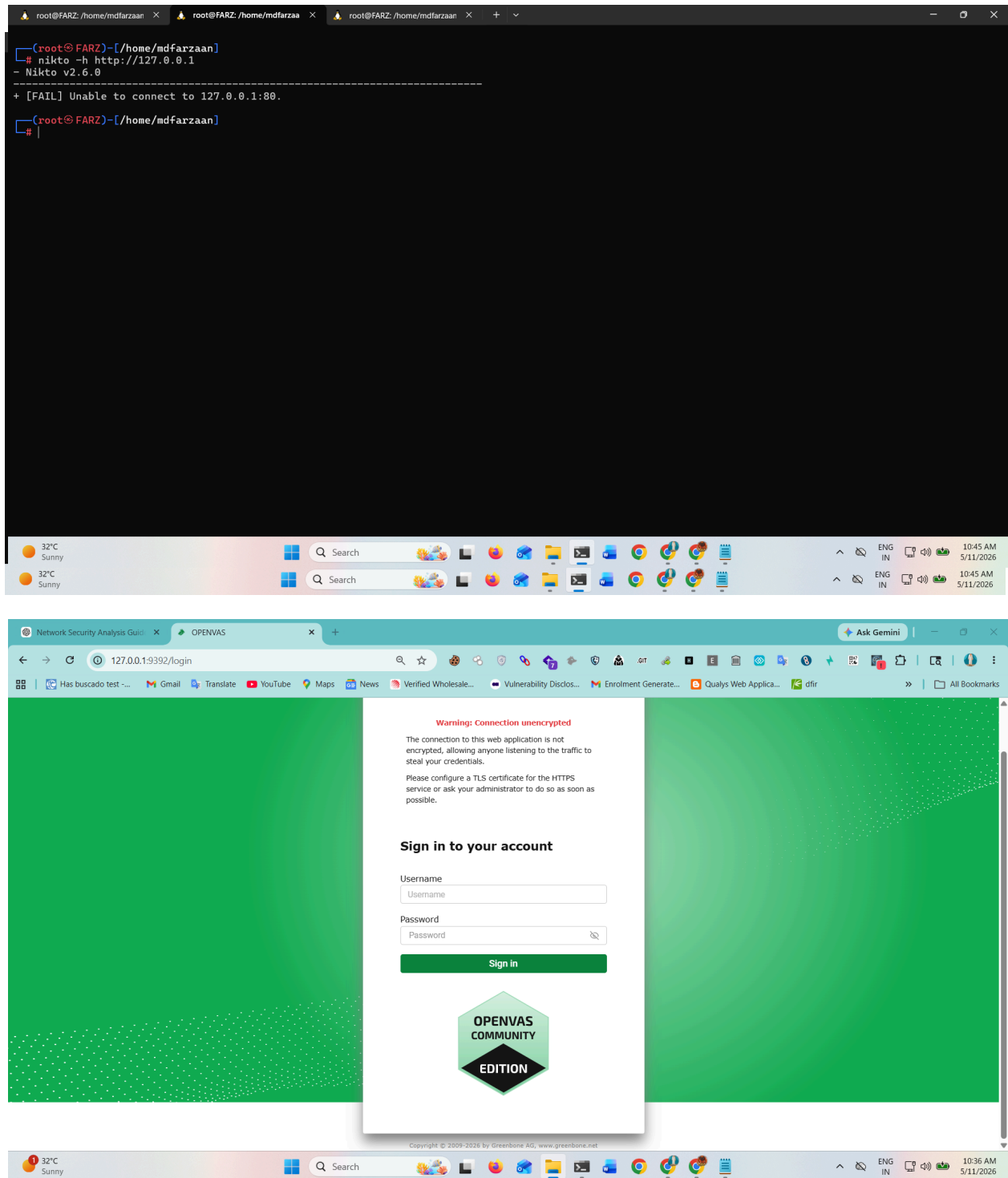
```
nikto -h http://127.0.0.1
```

The scans identified exposed services, running web servers, and potential outdated software components. The assessment demonstrated how vulnerability scanners help security teams identify weaknesses before attackers exploit them.

Example Findings:

- Outdated Apache Service
- Exposed HTTP Service
- Weak Web Configuration

Evidence:



4. DOCUMENTATION OF FINDINGS

All identified findings were documented in a structured format using spreadsheets and screenshots.

The following information was recorded during testing:

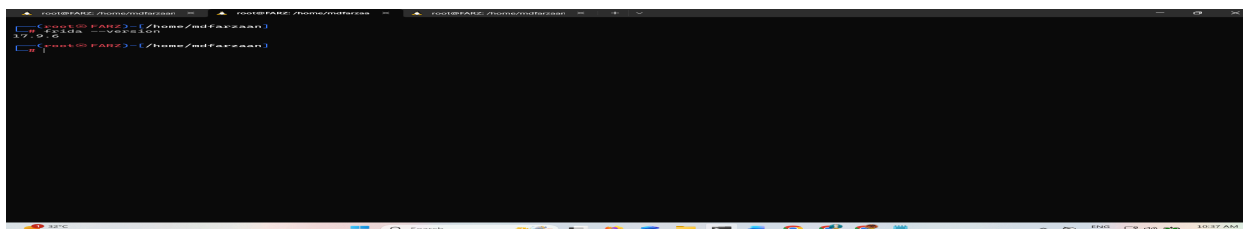
- IP addresses
- Open ports
- Running services
- Vulnerability descriptions
- Severity levels

Example Entry:

IP Address	Port	Service	Vulnerability
127.0.0.1	80	Apache	Outdated Version

The documentation process helped organize technical findings clearly and provided supporting evidence for reporting purposes.

Evidence:



```

root@FARZ: /home/mdfarzan
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1420 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:2c:7b:d1 brd ff:ff:ff:ff:ff:ff
    inet 172.26.100.1/20 brd 172.26.111.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::315:5dff:fe2c:7b01/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 0e:25:9f:23:48:12 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
root@FARZ: /home/mdfarzan

```

5. RISK ASSESSMENT

Risk assessment activities were performed using CVSS scoring and a basic likelihood-versus-impact matrix.

The CVSS Calculator was used to understand how vulnerabilities are categorized based on severity.

Example Severity Ratings:

- Critical
- High
- Medium
- Low

Example:

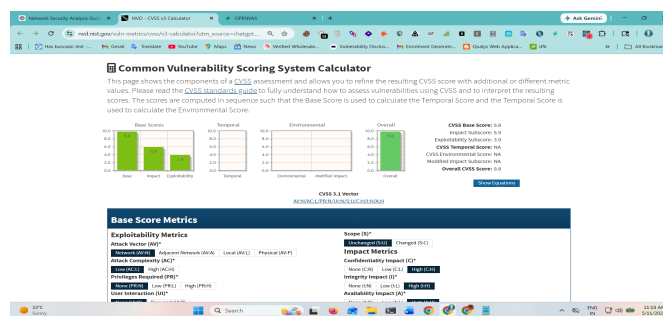
An SQL Injection vulnerability may receive a Critical severity score due to the possibility of unauthorized database access.

A simple 3x3 risk matrix was also used to prioritize findings based on:

- Likelihood of exploitation
- Potential impact on the system

This process helps organizations focus on fixing high-risk vulnerabilities first.

Evidence:



6. BASIC REPORTING

The final phase of the assessment involved preparing a professional vulnerability assessment report.

The report structure included:

- Executive Summary
- Technical Findings
- Screenshots and Evidence
- CVSS Severity Ratings
- Remediation Recommendations

Technical details and screenshots collected during testing were added to support each finding.

Example Recommendations:

- Update outdated software
- Apply security patches
- Restrict unnecessary services
- Harden server configurations
- Validate user input properly

The completed report was prepared in document format and exported as a PDF file for submission.

7. CONCLUSION

This practical lab provided hands-on understanding of vulnerability assessment fundamentals, including environment setup, vulnerability scanning, documentation, risk analysis, and reporting practices.

The activities demonstrated how cybersecurity professionals identify weaknesses, organize findings, assess risk severity, and recommend remediation steps using commonly used open-source security tools within a safe and controlled environment.

END OF REPORT